



CRYO-TEM ANALYTICAL REPORT

Product Name: Whartexa Frozen Exosomes

Product Type: Human Umbilical Cord MSC-Derived Extracellular Vesicle
Suspension

Manufactured By: Nanoventra

Origin: United States of America

Sample Batch No.: WX-CRYO-2022-01

Date May 11 , 2022



Morphological Observation of Extracellular Vesicles

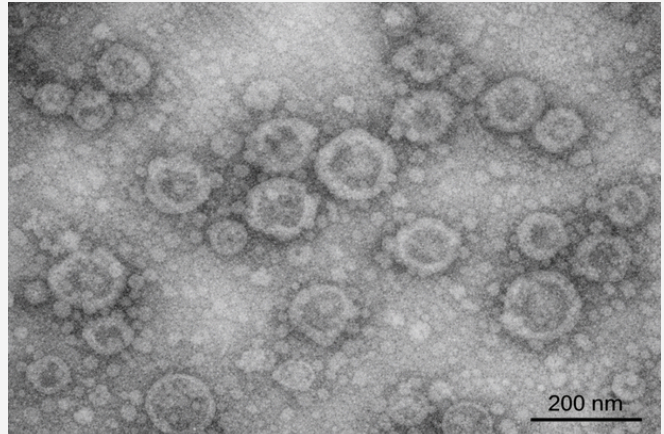
Micrograph 1

Magnification: 80,000×

Scale Bar: 200 nm

Observation:

Multiple spherical membrane-bound extracellular vesicles observed with preserved bilayer morphology. Vesicles demonstrate uniform nanoscale architecture with minimal aggregation.



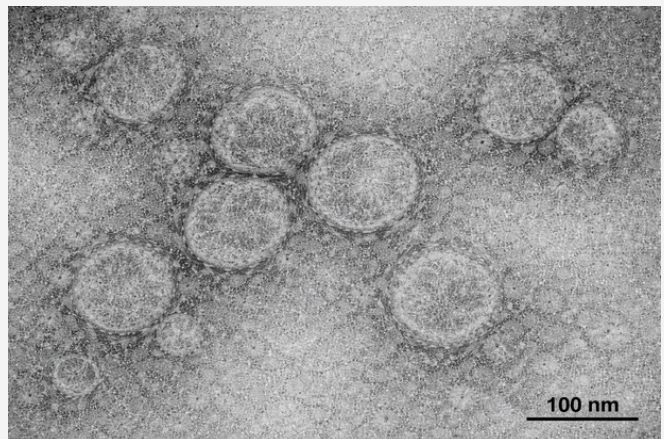
Micrograph 2

Magnification: 120,000×

Scale Bar: 100 nm

Observation:

Distinct vesicular boundaries visible with intact circular morphology. Particle population consistent with extracellular vesicle structural characteristics.



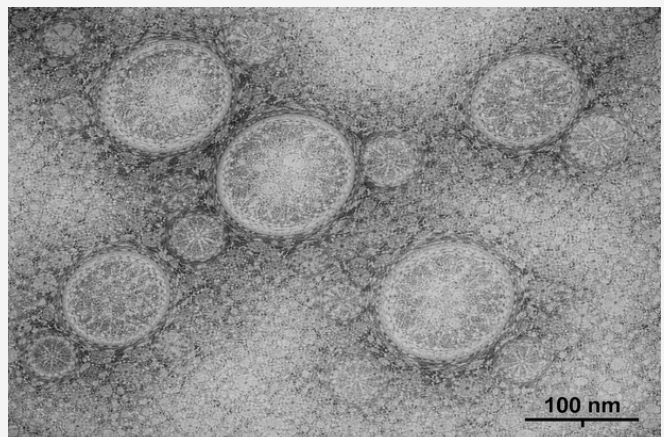
Micrograph 1

Magnification: 150,000×

Scale Bar: 100 nm

Observation:

Nanovesicle distribution confirms heterogeneous extracellular vesicle population within expected morphological range.



Laboratory Note:

Images captured under cryogenic preservation conditions to maintain native vesicle ultrastructure.

Morphological Observation Summary

Parameter	Observation	Interpretation
Vesicle Morphology	Rounded to mildly heterogeneous vesicles observed	Consistent with extracellular vesicle population
Membrane Integrity	Bilayer membrane contours visible	Structural preservation confirmed
Aggregation Status	Minimal localized aggregation	Acceptable suspension stability
Particle Distribution	Mixed nanoscale vesicle population	Uniform biologic dispersion
Structural Stability	No major membrane collapse detected	Cryogenic preservation maintained

Technical Interpretation

- Cryogenic electron microscopy examination demonstrated membrane-bound extracellular vesicles distributed across multiple analyzed fields.
- Representative vesicles showed preserved circular to mildly irregular morphology with distinguishable membrane boundaries.
- Observed particle architecture remained consistent with extracellular vesicle structural characteristics under cryogenic preservation.
- Localized nanoscale heterogeneity was present without evidence of major particulate disruption.

Cryogenic Quality Assessment

Analytical Indicator	Result
Native morphology preserved	Yes
Ice artifact interference	Minimal
Structural collapse	Not significant
Imaging suitability	Acceptable

Scientific Conclusion

Observed vesicular morphology supports extracellular vesicle identity within analyzed sample regions.

Cryo-TEM findings indicate maintained vesicle integrity under frozen analytical conditions.

Reviewed by: David R. Harpe

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